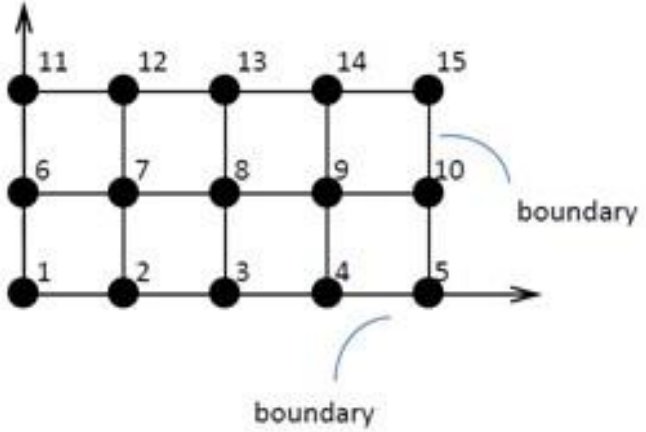
EE4047 – High Voltage Techniques I

New Topic:

Finite Difference Method

Week 12.1 – Monday Lesson

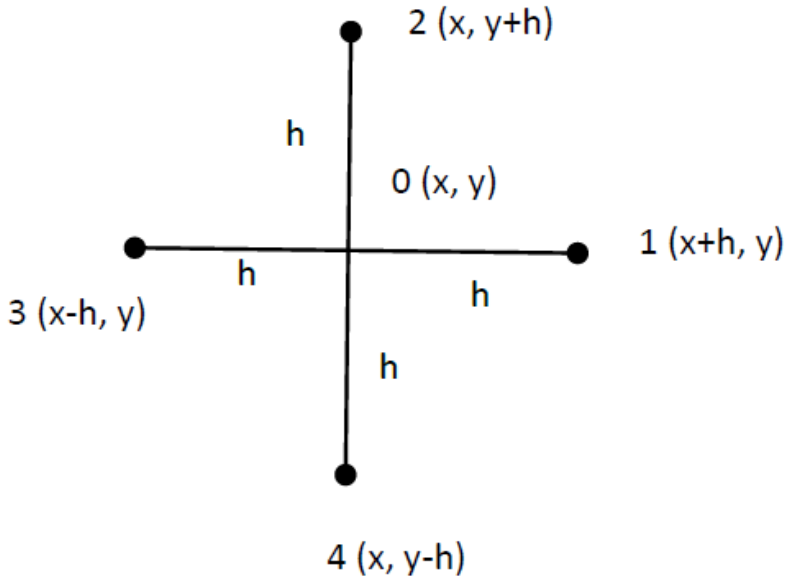


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FINITE DIFFERENCE METHOD

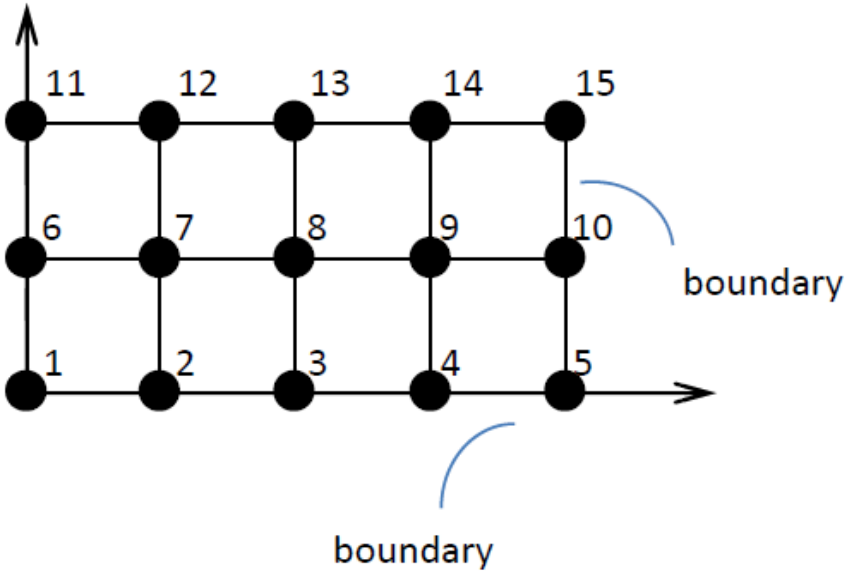
Finite Difference Method



$$V_0 = \frac{1}{4}(V_1 + V_2 + V_3 + V_4)$$

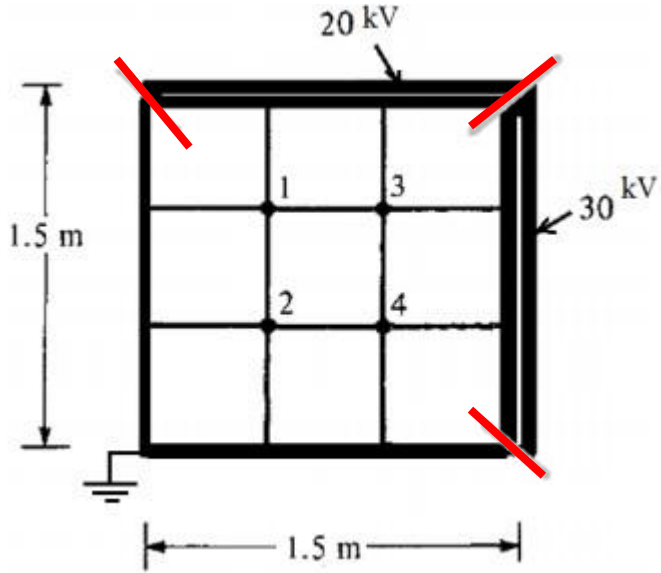
FINITE DIFFERENCE METHOD

Finite Difference Method



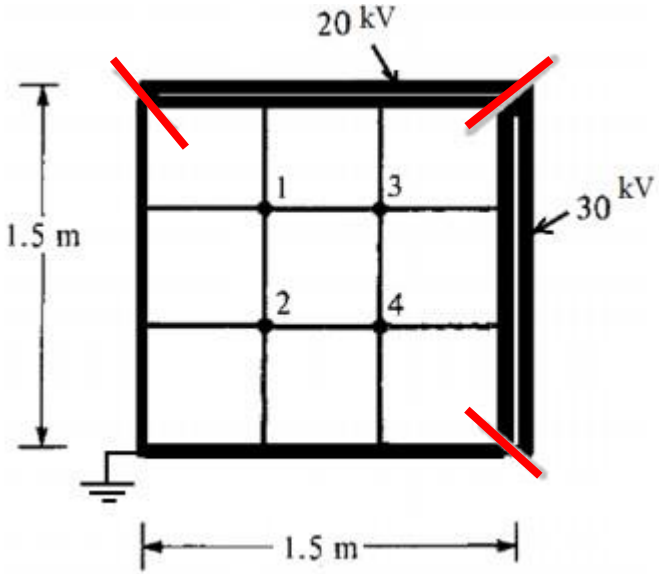
1 FINITE DIFFERENCE METHOD

Example-1: Determine the potential at the free nodes (1, 2, 3 and 4) in the figure using the finite difference method.



1 FINITE DIFFERENCE METHOD

Example-1: Determine the potential at the free nodes (1, 2, 3 and 4) in the figure using the finite difference method.



$$V_1 = \frac{1}{4}(V_2 + V_3 + 0 + 20) \longrightarrow 4V_1 = V_2 + V_3 + 20 \quad (I)$$

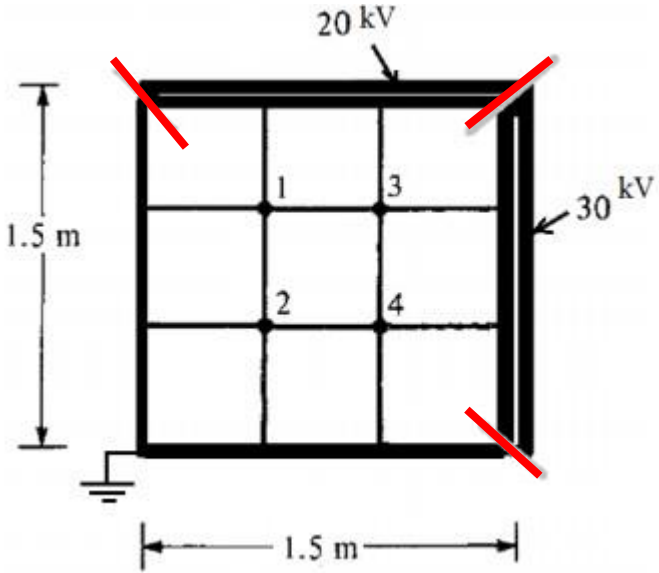
$$V_2 = \frac{1}{4}(V_1 + V_4 + 0 + 0) \longrightarrow 4V_2 = V_1 + V_4 \quad (II)$$

$$V_3 = \frac{1}{4}(V_1 + V_4 + 20 + 30) \longrightarrow 4V_3 = V_1 + V_4 + 50 \quad (III)$$

$$V_4 = \frac{1}{4}(V_2 + V_3 + 30 + 0) \longrightarrow 4V_4 = V_2 + V_3 + 30 \quad (IV)$$

1 FINITE DIFFERENCE METHOD

Example-1: Determine the potential at the free nodes (1, 2, 3 and 4) in the figure using the finite difference method.



(II) & (III) $\quad 4V_3 = 4V_2 + 50 \longrightarrow 4V_2 - 4V_3 = -50$ (V)

(I) & (II) & (IV) $\quad 4V_2 = \frac{1}{4}(V_2 + V_3 + 20) + \frac{1}{4}(V_2 + V_3 + 30)$

$16V_2 = 2V_2 + 2V_3 + 50 \longrightarrow -14V_2 + 2V_3 = -50$ (VI)

1 FINITE DIFFERENCE METHOD

Example-1: Determine the potential at the free nodes (1, 2, 3 and 4) in the figure using the finite difference method.

(V) & (VI)

$$\begin{cases} 4V_2 - 4V_3 = -50 \\ 2\times\ -14V_2 + 2V_3 = -50 \end{cases} \rightarrow \begin{cases} 4V_2 - 4V_3 = -50 \\ -28V_2 + 4V_3 = -100 \end{cases}$$

$$V_2 = \frac{-150}{-24} \Rightarrow V_2 = 6.25 \text{ kV}$$

(V) →

$$V_3 = \frac{-50 - 4V_2}{-4} \rightarrow V_3 = \frac{-50 - 4(6.25)}{-4} \rightarrow V_3 = 18.75 \text{ kV}$$

(I) →

$$V_1 = \frac{V_2 + V_3 + 20}{4} = \frac{6.25 + 18.75 + 20}{4} \rightarrow V_1 = 11.25 \text{ kV}$$

(IV) →

$$V_4 = \frac{V_2 + V_3 + 30}{4} = \frac{6.25 + 18.75 + 30}{4} \rightarrow V_4 = 13.75 \text{ kV}$$

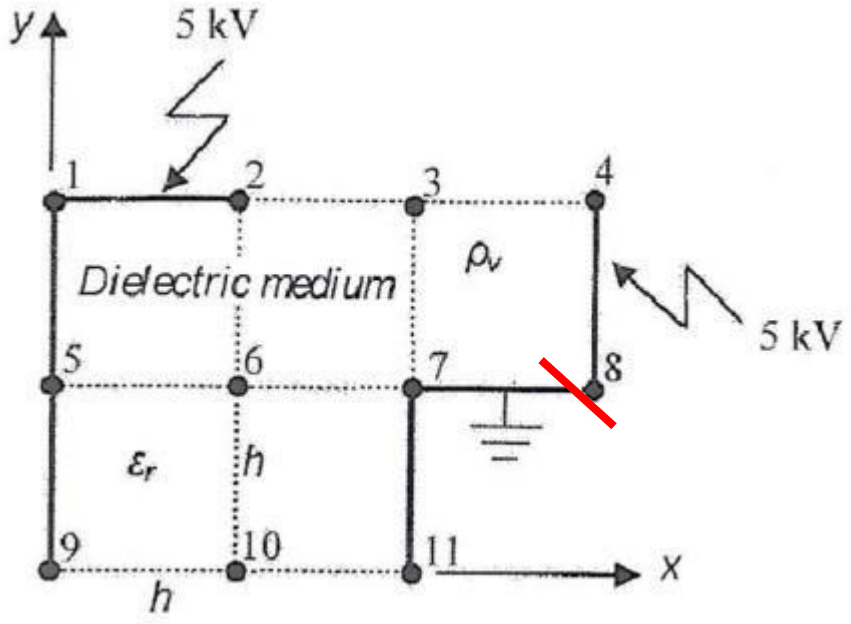
FINITE DIFFERENCE METHOD

Example-2: Determine the potential at the free nodes (3, 6 and 10) in the figure using the finite difference method.

Dirichlet boundary conditions:

$V_1=V_2=V_4=V_5=V_8=V_9=5$ kV, $V_7=V_{11}=0$

Neumann boundary conditions: $(\frac{\partial V}{\partial y})_3 = (\frac{\partial V}{\partial y})_{10} = 0$



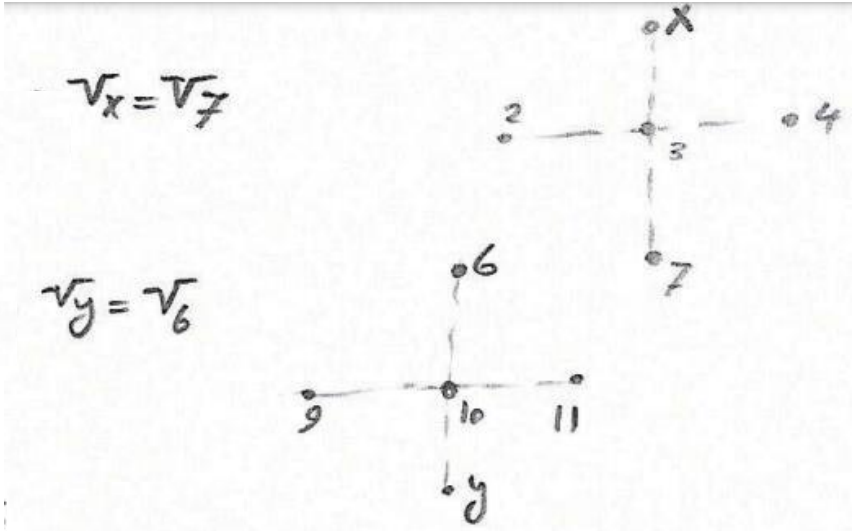
FINITE DIFFERENCE METHOD

Example-2: Determine the potential at the free nodes (3, 6 and 10) in the figure using the finite difference method.

Dirichlet boundary conditions:

$V_1=V_2=V_4=V_5=V_8=V_9=5\ ^{kV}, \quad V_7=V_{11}=0$

Neumann boundary conditions: $(\frac{\partial V}{\partial y})_3 = (\frac{\partial V}{\partial y})_{10} = 0$



$V_x = V_7$

$V_y = V_6$

$V_3 = \frac{1}{4}(V_2 + V_4 + 2V_7) = \frac{1}{4}(5+5+0) \Rightarrow$

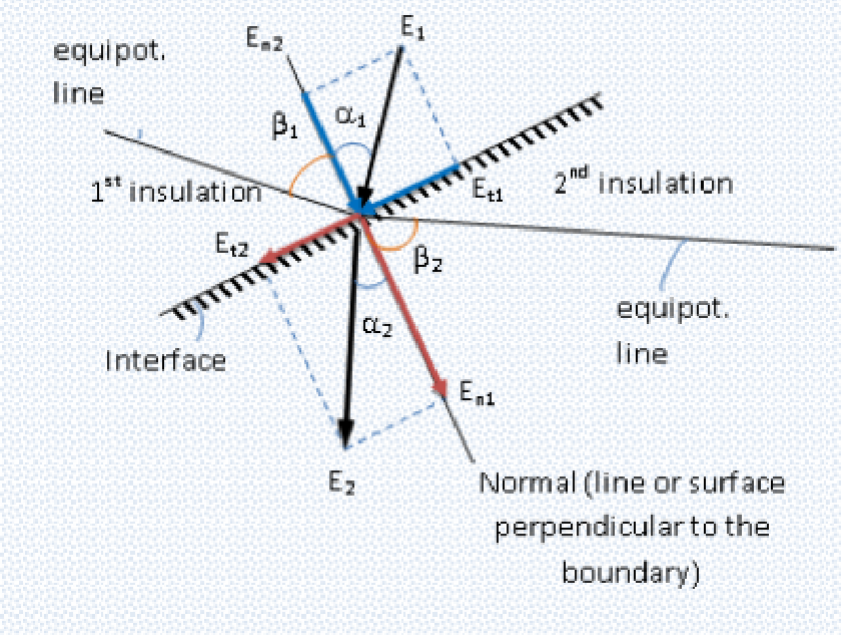
$V_3 = 2.5\ ^{kV}$

EE4047 – High Voltage Techniques I

New Topic:

Refraction of Electric Fields

Week 12.1 – Monday Lesson



Asst. Prof. Dr. M. Alparslan

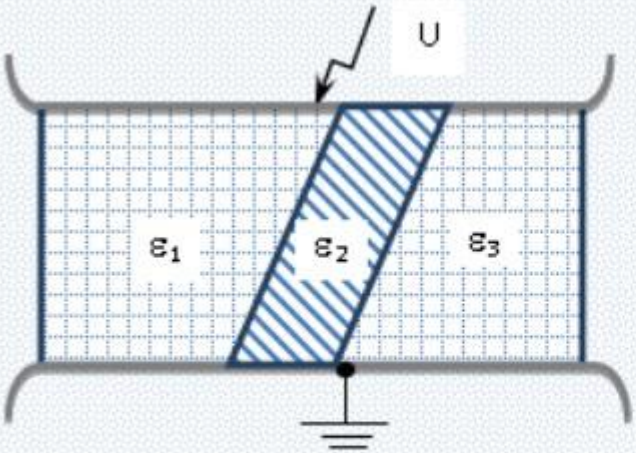
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OUTLINE

1 Introduction

2 Refraction

Non-parallel configuration

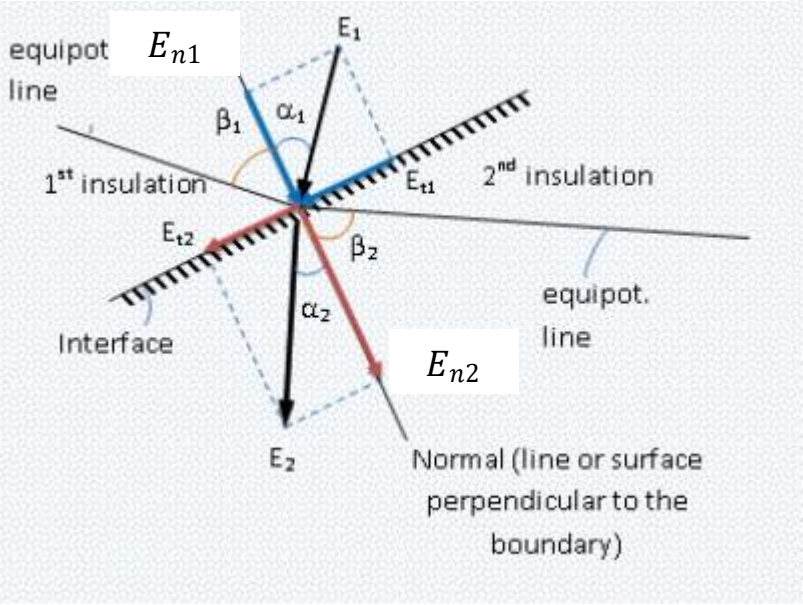


What happens to electric field if the plates are NOT parallel?

In order to evaluate that, we have to know the behaviour of field lines at boundary surfaces.

2 REFRACTION

Refraction of electric field lines and equipotential lines



- α_1, α_2: Refraction angles of electric field lines.
- β_1, β_2: Refraction angles of equipotential lines

(REMEMBER: always perpendicular to the field lines!!!)

- E_{t1}, E_{t2}: tangent component of the electric fields E_1 and E_2
- E_{n1}, E_{n2}: normal (perpendicular) component of the electric fields E_1 and E_2

$$E_{t1} = E_{t2}$$

END OF THE LESSON

Any questions?